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Best practices for source-based research on misinformation and news trustworthiness

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Researchers need reliable and valid tools to identify cases of untrustworthy information when studying the spread of misinformation on digital platforms. A common approach is to assess the trustworthiness of sources rather than individual pieces of content. One of the most widely used and comprehensive databases for source trustworthiness ratings is provided by NewsGuard. Since creating the database in 2019, NewsGuard has continually added new sources

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and reassessed existing ones. While NewsGuard initially focused only on the US, the database has expanded to include sources from other countries, such as Germany and Italy. In addition to trustworthiness ratings, the NewsGuard database contains various contextual assessments of the sources, which are less often used in contemporary research on misinformation. In this work, we provide an analysis of the content of the NewsGuard database, focusing on the temporal stability and completeness of its ratings across countries, as well as the usefulness of information on political orientation and topics for misinformation studies. We find that the trustworthiness ratings are temporally stable and that coverage seems complete for the US and Germany. Information on the political orientation and topic labels of sources are relatively complete and can be valuable assets in characterizing sources beyond trustworthiness. By evaluating the database over time and across countries, we identify potential pitfalls that threaten the validity of using NewsGuard as a tool for quantifying untrustworthy information. Lastly, we provide recommendations for digital media research on how to avoid these pitfalls and discuss appropriate use cases for the NewsGuard database and source-level approaches in general.

Keywords: misinformation, source trustworthiness, NewsGuard

Misinformation spreads on both social media platforms (Lazer et al., 2018) and mainstream media channels (Tsfati et al., 2020). However, only a fraction of news consumption is attributed to false news — roughly 0.15% (Allen et al., 2020) — and only a minority of users actively engage with unreliable information online (Baribi-Bartov et al., 2024; Grinberg et al., 2019). Typically, these individuals hold strong political or ideological beliefs and accept and actively look for information that echoes their opinions (Ecker et al., 2022) — information that often comes from political elites or established media sources (Tsfati et al., 2020). Misinformation capitalizes on these dynamics to stoke conflict and evoke negative emotions, thereby exacerbating partisan divides and fueling inter-group hostility (González-Bailón et al., 2023; Robertson et al., 2023). Against a backdrop of declining trust in both media outlets (Newman et al., 2023) and institutions (Bennett and Livingston, 2020), it becomes increasingly crucial to comprehend the mechanisms driving the dissemination of misinformation, particularly in the digital sphere. While digital media facilitates the measurement of these dynamics, it also has the potential to reinforce polarization and misinformation (Lorenz-Spreen et al., 2022).

Scholars across various disciplines have extensively studied the phenomenon of misinformation, employing terms such as misinformation, disinformation, fake news, rumors, or conspiracy theories, often used interchangeably (Guess and Lyons, 2020). Although these terms share a focus on veracity, they differ in terms of intentionality. Misinformation is commonly used as an umbrella term for inaccurate or unreliable information, regardless of intent, e.g., Ecker et al. (2022) or van der Linden (2022). However, the ambiguous nature of misinformation presents significant methodological challenges, granting researchers several degrees of freedom. Defining the boundaries of misinformation—essentially determining where the truth ends—is a complex task for several reasons. First, misinformation often appears in subtle forms rather than obvious lies (Allen et al., 2024; Altay et al., 2023; Bakir and McStay, 2018; Vargo et al., 2018), making it hard to define where it starts. Implicit, often partisan, interpretations of facts differ from explicit lies, making subtle forms of misinformation hard to detect. Second, identifying misinformation involves ethical considerations, with different experts reaching different conclusions. To navigate these challenges, researchers often focus on binary (true/false) cases of misinformation, typically identified by fact-checkers. While this approach addresses some ethical concerns by clearly classifying false information, it captures only a small portion of misinformation. Finally, conceptual choices are often shaped by methodological constraints (e.g., available data) or disciplinary preferences (e.g., methodological traditions), which is why misinformation is often used as a general umbrella term for different phenomena (Weeks and Gil de Zúñiga, 2021). Consequently, this semantic and methodological variability has contributed to imprecise or inconclusive findings regarding the prevalence and impact of misinformation. While some scholars, such as Altay et al. (2023) or Budak et al. (2024), argue that the effects of misinformation are overstated, calling it a symptom of broader social issues, others, like Tay et al. (2024), reason that the heterogeneity in measuring misinformation contributes to divergent evidence and creates a false dichotomy between understanding misinformation as a symptom or a cause (also see Bozarth et al., 2020).

In digital media research, one of the most prominent ways to measure misinformation has been to use a list of sources that are known to share misinformation frequently. NewsGuard has emerged as the most widely used database for source ratings (Celadin et al., 2023; Guess et al., 2020; Lasser et al., 2022; Pratelli et al., 2023; Robertson et al., 2023) and is especially popular in misinformation research for its comprehensive source-level trustworthiness scores, which are compiled based on web tracking data and rated and updated by professional editors and journalists In addition to its comprehensiveness, NewsGuard also offers a more fine-grained assessment of the trustworthiness of sources based on adherence to journalistic quality criteria and provides a point score ranging from 0 (the source is untrustworthy because it severely violates basic journalistic standards) to 100 (the source adheres to all standards of credibility and transparency). Since the start of the database in 2019, NewsGuard has constantly added new sources. While initially the focus of NewsGuard was the US, the database has expanded to other countries, such as Germany and France. Given the popularity of the NewsGuard database as a tool to quantify the prevalence of misinformation, an in-depth analysis of the temporal stability of ratings and the completeness of the database for different countries is warranted. Furthermore, NewsGuard requires a license, and researchers in need of a measurement instrument to quantify misinformation must weigh the advantages and disadvantages of source-based methods and the investment required to access the NewsGuard database. With the present in-depth analysis of the database's content we aim to provide researchers with the necessary information to make this decision. Where warranted, we provide recommendations for effectively utilizing the NewsGuard database and source-based approaches alongside our analysis of the content of the database.

Tracking Misinformation on Digital Media

To study the diffusion or discussion of misinformation on digital platforms, computational researchers commonly opt for either content- or source-based methodologies, searching for web links to content (via the URL) or a source (via their web domain). Both approaches rely on experts, such as fact-checkers, to curate lists of content or sources, which serve as the foundation for collecting data from digital platforms. Source-based strategies, which are the focus of the present work, offer the advantage of encompassing a broader spectrum of sources and narratives, thus enriching the sample by considering a more diverse information environment. Source-based approaches also enable the inclusion of subtler forms of misinformation and facilitate accurate estimations of both the scale (Allen et al., 2020; Grinberg et al., 2019; Guess et al., 2020; Yang et al., 2021) and dynamics of dissemination (Lasser et al., 2022, 2023; Pennycook et al., 2021; Robertson et al., 2023; Shao et al., 2018).

The precision of source-based approaches hinges mainly on the selection and quality of source ratings within a given list. Strictly speaking, this conceptualization covers information disseminated by unreliable sources (rather than individual pieces of misinformation). Therefore, observed patterns may be intricately linked to the inherent biases or external forces in the selection of sources. For instance, if we analyzed misinformation in politicians' social media posts, an unbalanced selection of unreliable sources would most likely reveal patterns due to partisanship rather than patterns universal to misinformation. This underscores the importance of assessing confounding factors such as political biases or editorial practices of the sources, considering that a lack of editorial oversight can compromise the quality of news reporting (Lazer et al., 2018). In other words, to obtain a balanced selection of sources and account for the influence of source-related variables other than reliability, source-based approaches can incorporate assessments of political, cultural, or editorial biases and misleading tendencies while maintaining generalizability. Nonetheless, several problems remain in practice: First, source categorizations often rely on a binary classification system (fake/real), oversimplifying the nuanced spectrum of misinformation. Second, it is crucial to acknowledge that source ratings vary across time and cultural contexts, for instance, following a change in ownership or editorial practices. For example, Lin et al. (2024) combined source-level quality ratings with context-specific keywords, e.g., hashtags linked to an anti-vaccination protest in Ottawa to account for misinformation-sharing in different populations. Third, while source judgments are typically made without requiring specific topic expertise, they often reflect the journalistic traditions of a particular media ecosystem, which may not be universally applicable across different contexts (Brüggemann et al., 2014). Source-based approaches are a promising way to track misinformation online at scale. However, to study the spread of unreliable news, scientists need access to reliable and credible databases of source ratings.

NewsGuard: A Popular Database for Source Ratings

Since the 2016 US elections, the most popular way to track misinformation online has been to use a list of untrustworthy or unreliable sources and search for their web domains in URL text (Lasser et al., 2022; Grinberg et al., 2019; Guess et al., 2020; Shao et al., 2018; Yang et al., 2021). The organization NewsGuard offers the most comprehensive list of such domains (Lin et al., 2023). It does not just provide a blacklist of untrustworthy sources but provides a trustworthiness score for each source, covering news sources across the entire spectrum of news quality. A trained team of experts, mainly consisting of journalists and editors, rates the news sources based on nine journalistic quality criteria[1] and regularly updates both the coverage of the list by adding new sources and the trustworthiness ratings of existing sources. Since its start in 2019, the database has been constantly growing and has recently extended to countries beyond the United States, such as Germany and Italy. With recent political events and crises, demand for such databases has increased, enhancing the popularity of NewsGuard. For instance, it has been used to study user engagement with unreliable news during the 2016 and 2020 US elections (Pratelli et al., 2023; Robertson et al., 2023), to assess the changing quality of cited information of political elites in the US (Lasser et al., 2022) or alternative understandings of honesty and truth in political discourse (Lasser et al., 2023), and to estimate the effects of source labeling on decreasing misinformation sharing (Celadin et al., 2023).

However, the NewsGuard database was initially created as a tool for brand security rather than scientific research. In addition, employing journalists and editors to assess the sources included in the list incurs significant expenses. Given its predominantly private audience and expenses, the NewsGuard database is only available to subscribers who pay to license the database. Here, we provide a comprehensive assessment of the database to help misinformation researchers weigh the resources required to access the NewsGuard database and the advantages and limitations of source-based approaches in general. We aim to answer the following main research question: How volatile are source-level trustworthiness ratings, such as those in the NewsGuard database, across time and language contexts? To answer our research question, we first assess the distribution of trustworthiness ratings, the process

[1]https://www.newsguardtech.com/ratings/rating-process-criteria/

of updating them, and the inclusion and removal of sources over time. Second, we assess the completeness of the database for different countries. Lastly, we estimate the usefulness of other source-level information provided in the database, such as the political orientation and topics covered by sources. To assess different aspects of the database, we introduce three sub-questions:

How stable are trustworthiness ratings over time? Investigating the development of the NewsGuard database over time is important to estimate whether using an older version of the database distorts results and should be used to measure trustworthiness in older data. We describe the composition of the trustworthiness score and different database versions. We also explore the frequency and reasons for major changes, such as how often ratings change or sources are removed. Additionally, we provide an exemplary reproduction of published research using NewsGuard ratings from different points in time to assess the degree to which results vary based on changes in the measurement instrument.

How complete is the database across countries, particularly for non-US countries? A recent study has shown high agreement of NewsGuard with other expert-rated lists in the US context (Lin et al., 2023), but an assessment for other countries is missing. Therefore, we describe the coverage across and differences between countries. Further, we compare the completeness of information provided by NewsGuard across different contexts by cross-checking the NewsGuard database with a dataset of domains shared on social media for the US, Germany, and UK, and other existing lists of low-quality web domains.

How valuable are contextual source labels for misinformation research? NewsGuard labels the topics covered as well as the political orientation sources. We analyze the temporal stability and completeness of these labels across countries. Additionally, we manually validate the labels for German-speaking sources. We assess in what way such labels can complement source characterizations and shed light on aspects of misinformation beyond trustworthiness.

Dataset Construction

NewsGuard curates the list of sources it rates by tracking online activity. On their website, NewsGuard claims to have reviewed news sources accounting for 95% of online engagement, including news consumed and shared online. The organization also manually adds sources they deem influential or that changed their web domain. The database automatically updates every hour and allows tracking changes back to March 2019, when the organization started operating. Still, how domains are selected or removed is not disclosed on the website. As of September 15, 2024, the database has 12,288 entries in total, some without a trustworthiness rating (7.6%). This includes 70 sources classified as a "platform" (e.g., YouTube), 63 sources judged as "satire" (e.g., The Onion) and 806 lifestyle sources (e.g., healthquote- free.com). For the remaining 11,349 sources, the database includes the following characteristics:

- Unique identifier for each rating and date of the last update
- Domain name and parent domain (e.g., nytimes.com)
- A trustworthiness rating as a quasi-continuous scale (between 0 and 100) and binary labels (N = Not Trustworthy/T = Trustworthy) based on a threshold at 60
- Language and country associated with the source
- Political orientation of the source (e.g., "left" and "right")
- Topics covered (e.g., "local news", "political news", "health information") by the source

The team of NewsGuard experts regularly judges the trustworthiness of the selected news sources based on nine journalistic criteria that are universally applied across countries. The trustworthiness score is a composite score of these nine indicators. More precisely, experts assign a binary label for each criterion. Each criterion has a weight, a number of points, which together make up the overall trustworthiness rating (ranging from 0 to 100). Here, we list the criteria and their weights in descending order[2]:

[2]https://www.newsguardtech.com/ratings/rating-process-criteria/

1. “Does not repeatedly publish false content”, changed to “Does not repeatedly publish false or egregiously misleading content” in December 2023 (22 points)
2. “Gathers and presents information responsibly” (18 points)
3. “Regularly corrects or clarifies errors”, changed to “Has effective practices for correcting errors” in December 2023 (12.5 points)
4. “Handles the difference between news and opinion responsibly” (12.5 points)
5. “Avoids deceptive headlines” (10 points)
6. “Website discloses ownership and financing’ (7.5 points)
7. “Clearly labels advertising” (7.5 points)
8. “Reveals who’s in charge, including possible conflicts of interest” (5 points)
9. “The site provides the names of content creators, along with either contact or biographical information” (5 points)

Since the overall score is a sum of the individual scores for each criterion, it is not truly continuous, as some values (for example, 1, 2, 3, or 4 points) cannot be achieved by any combination of criteria.

Descriptive Analyses

How stable are trustworthiness ratings over time?

In the following, we review how the database, particularly the trustworthiness ratings and source selection, have changed over the years since the first version of the database was compiled in 2019.

Coverage across time. Over the span of five years, NewsGuard has grown from 2,375 to 12,288 entries in total, as shown in Figure 1A to C. NewsGuard regularly adds new sources but rarely removes any, i.e., the database has been growing to four times its initial size. When counting the added domains based on monthly comparisons of the list, NewsGuard has added 8,906 domains over time but only removed 685, adding an average of 137 new domains per month.

NewsGuard ratings are available for customers in an online database (Amazon S3 bucket), where a current snapshot is uploaded every hour. We sampled a snapshot from the first hour of the 15th day of every month. When we instead sampled from the second day of each month, we observed two irregular dips in the number of sources (see May 2022 and February 2024 in Figure 1A). It is likely that errors caused this fluctuation. For instance, in May 2022, the approximately 3,500 removed sources that did not appear in the database were present in the database two weeks later. Yet, this sudden disappearance and reappearance of sources illustrate the overall volatility of the database. We therefore advise researchers to inspect several database snapshots and check their consistency before choosing one.

The most recent snapshot contains ratings for 11,349 unique domains in the 12,288 total entries; indicating that some sources appear multiple times. Typically, these sources publish in different languages, i.e., they have the same web domain but provide multilingual content, or they have an entry as a local and a global source. For instance, the Austrian website servustv.com produces both German- and English-language output and appears with the country label "AT" and the label "ALL". According to NewsGuard, the latter serves as a generic label for the English-language translation of sites that were also rated in their respective language. In other words, each site with a label "ALL" also exists in at least one other language, and the label "ALL" can largely be ignored. In line with that, duplicated sites have identical trustworthiness ratings across country or language categories, except for only three cases. Therefore, dropping duplicated domain entries does not substantially affect downstream research results. However, if a classification of sources by country is necessary for the research question at hand, researchers should keep duplicated domains.

Furthermore, 5,754 of the rated domains belong to only 444 unique parent domains. Some parent domains have multiple variations of the domain name, probably representing the same source, e.g., theragingpatriot.org has domains ending with .blog, .com, .net, and .pro. Other parent domains have very different sub-domains, e.g., 24usnews.com and africdaily.com belong to charmdaily.com. The other 6,534 ratings do not have a parent domain entry in the database. Domains belonging to the same parent domain usually have the same trustworthiness rating. Furthermore, NewsGuard updates domains belonging to the same parent domain at the same time, and they seem to assign the same unique identifier to those domains. Therefore, while the database contains 10,862 unique domains, it only contains 7,073 unique identifiers and, therefore, individually rated (parent) domains.

Distribution. NewsGuard trustworthiness ratings range from 0 to 100. Historically, NewsGuard considered sources with a rating below 60 as not trustworthy, therefore assigning binary trustworthiness labels[3]. Since February 2023[4], NewsGuard uses a more granular categorization, classifying sources with a score of 100 as ”high credibility”, scores between 75 and 99 as ”generally credible”, scores between 60 and 74 as ”credible with exceptions”, scores between 40 and 59 as ”proceed with caution”, and scores between 0 and 39 as ”proceed with maximum caution”. Notably, NewsGuard is not a blacklist. In fact, close to 60% of sources have an overall rating of over 60 (see Fig. 1B for the distribution of trustworthiness scores in the most recent vs. the first version of the database and Fig. A.1 for distributions per year). Furthermore, the distributions also clearly show a multimodal structure, indicative of the quasi-continuous nature of the overall NewsGuard score. In addition, some combinations of characteristics seem much more prevalent than others, indicating that the individual characteristics are not independent (see also “Composition of trustworthiness score” below).

When looking at the score distribution over time, the skew towards high ratings diminishes slightly: while the average score was 71.8 (SD=33.3) in 2019, it is now at 63.6 (SD=32.6) as shown in Figure 1C. Various explanations for decreases in trustworthiness over time are possible: First, the database has tripled in size and added many sources with lower trustworthiness scores (on average 59.6, SD=27.5).

[3]Snapshot of the described rating process from December 31st, 2022: https://www.newsguardtech.com/ratings/rating-process-criteria/
[4]Snapshot of the updated rating process from February 1st, 2023: https://www.newsguardtech.com/ratings/rating-process-criteria/

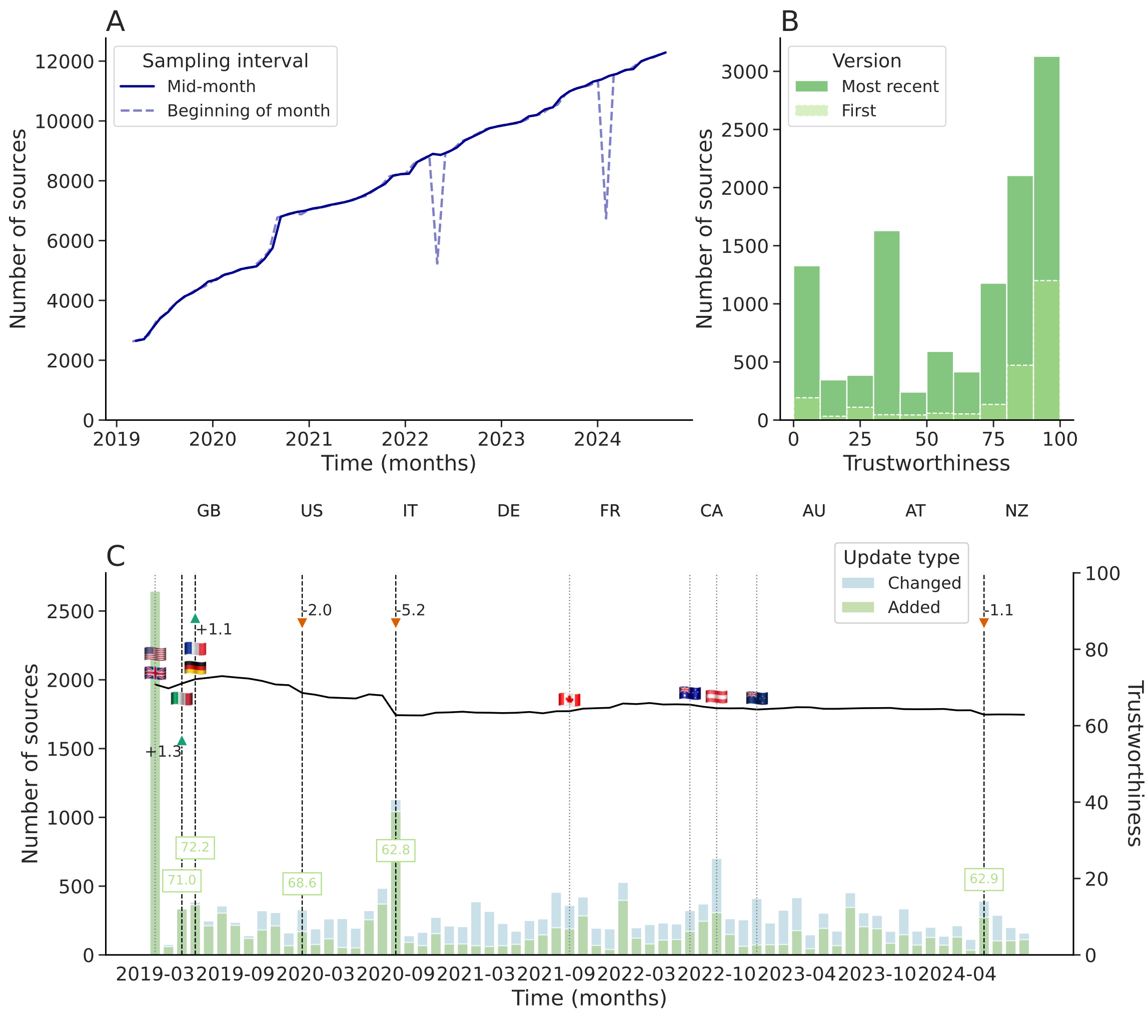

Figure 1. Description of trustworthiness ratings with A showing the number of sources with rating over time based on two different sampling strategies: when sampling one snapshot of the database mid-month (solid line), we observe a steady increase, whereas sampling at the beginning of the month results in two irregular dips (dashed line). Panel B: distribution of trustworthiness in the first and most recent database as a histogram. Panel C: changes in trustworthiness scores over time (monthly granularity), with flags (respective country codes in order of appearance) and dotted lines indicating when countries were added and vertical lines highlighting the top five updates (i.e., major score changes, also shown with arrows). The height of the bars describes the number of sources added vs. changed, with colors indicating the proportions. Note: We include the average trustworthiness of sources added in green for the major changes. Countries are listed in the order added at the top of panel C.

A significant number of sources with lower ratings were added, especially during the outbreak of the COVID-19 pandemic in 2020. The dashed lines with arrows in Figure 1C show the top five updates in terms of their impact on the average trustworthiness. The height of the bars shows the number of sources added vs. changed (with proportions in colors and the average trustworthiness of added sources in green). For all major updates, the proportion of added sources is larger than changed sources. The highest number of sources added was in September 2020, when 1,030 sources with on average very low trustworthiness (M=35.3, SD=14.1) were included. It is unclear if this development is due to an effort by NewsGuard to include more untrustworthy news or a global increase in untrustworthy sources.

Second, NewsGuard employs teams of experts per country, potentially leading to differences in coverage and ratings between countries. Dips in trustworthiness may be related to country-specific sources being added. For instance, 1,015 of the 1,030 added sources in September 2020 are coded as US, resulting in a drop in trustworthiness of roughly 5%. In contrast, adding entirely new countries to the database seems to have little effect on the overall trustworthiness score (see the dotted lines with country flags in Figure 1C), indicating that NewsGuard curates a balanced mix of trustworthy and untrustworthy sources when adding a new country. Overall, smaller dips appear across all countries covered and are most likely related to many low-quality sources gaining traction around the time of the COVID-19 pandemic.

A third influence on the overall trustworthiness score is updates of existing ratings. NewsGuard indicates the date when a rating was last updated, even if the score did not change. Updates to entries are documented on average once every eight months (249 days), but this varies strongly per source between a few seconds to a maximum of 790 days (interquartile range: 157 to 334 days). When updated, trustworthiness scores mostly remain unchanged (on average, 20% of updates mean a change in the rating). In other words, ratings realistically change every two years (M=570 days, IQR=397 days to 670 days). In addition, the majority of sources have not decreased in trustworthiness (average change of rating per source=0.1, SD=3.5). However, a few sources drastically dropped in trustworthiness; for instance, conservativedailynews.com lost 80 points in one update in November 2020. Figure A.2 in the appendix illustrates the differences in consecutive scores over time, with major score updates in early 2020.

Overall, the decrease in the overall trustworthiness of sources we can see in Figure 1C can, therefore, be attributed to NewsGuard adding untrustworthy sources rather than previously trustworthy sources losing points. This interpretation is also supported by our analysis of changes in individual rating criteria below.

Composition of trustworthiness score. The trustworthiness score is a composite score calculated based on nine journalistic criteria. For each source, NewsGuard's raters assign a Boolean value (yes/no) per criterion. The trustworthiness score is then calculated as the sum of the weighted criteria. Further explanation of the criteria and their weights can be found in the NewsGuard FAQ[5].

In the latest version of the dataset, 81.3% of sources fulfill the criterion "Avoids deceptive headlines" (9,217 sources). The second most prevalent criterion is "Does not repeatedly publish false or egregiously misleading content", which is met by 81.2% of all sources. These two top criteria often go together: When one is satisfied, the other tends to be as well (Pearson's correlation coefficient = 0.9). A factor analysis with oblique rotation confirms that the two criteria load on one factor, accounting for 58.1% of the explained variance, with factor loadings > 0.8. All other criteria share 36.9% of the variance explained (factor loadings > 0.5). In line with that, 11.8% of all sources meet only the two top criteria combined, while 13.4% of sources satisfy none of the nine journalistic criteria. The most common combination of criteria is that all nine criteria are satisfied simultaneously (13.5% of sources). We provide a correlation matrix of criteria in the Appendix (Table A.1).

We find that around 60% of sources have never changed in their fulfillment of journalistic quality criteria during updates. We present update patterns of individual criteria over time in Figure 2A. The most common change is sources stopping disclosing ownership and financing, accounting for 12.3% of the observed changes. The opposite change (a disclosure of ownership and financing) is the third most common pattern, occurring in 10.7% of score updates. This indicates either that NewsGuard frequently checks criteria related to ownership or that adherence to such criteria is more volatile.

[5]From December 31st, 2023: https://www.newsguardtech.com/ratings/rating-process-criteria/

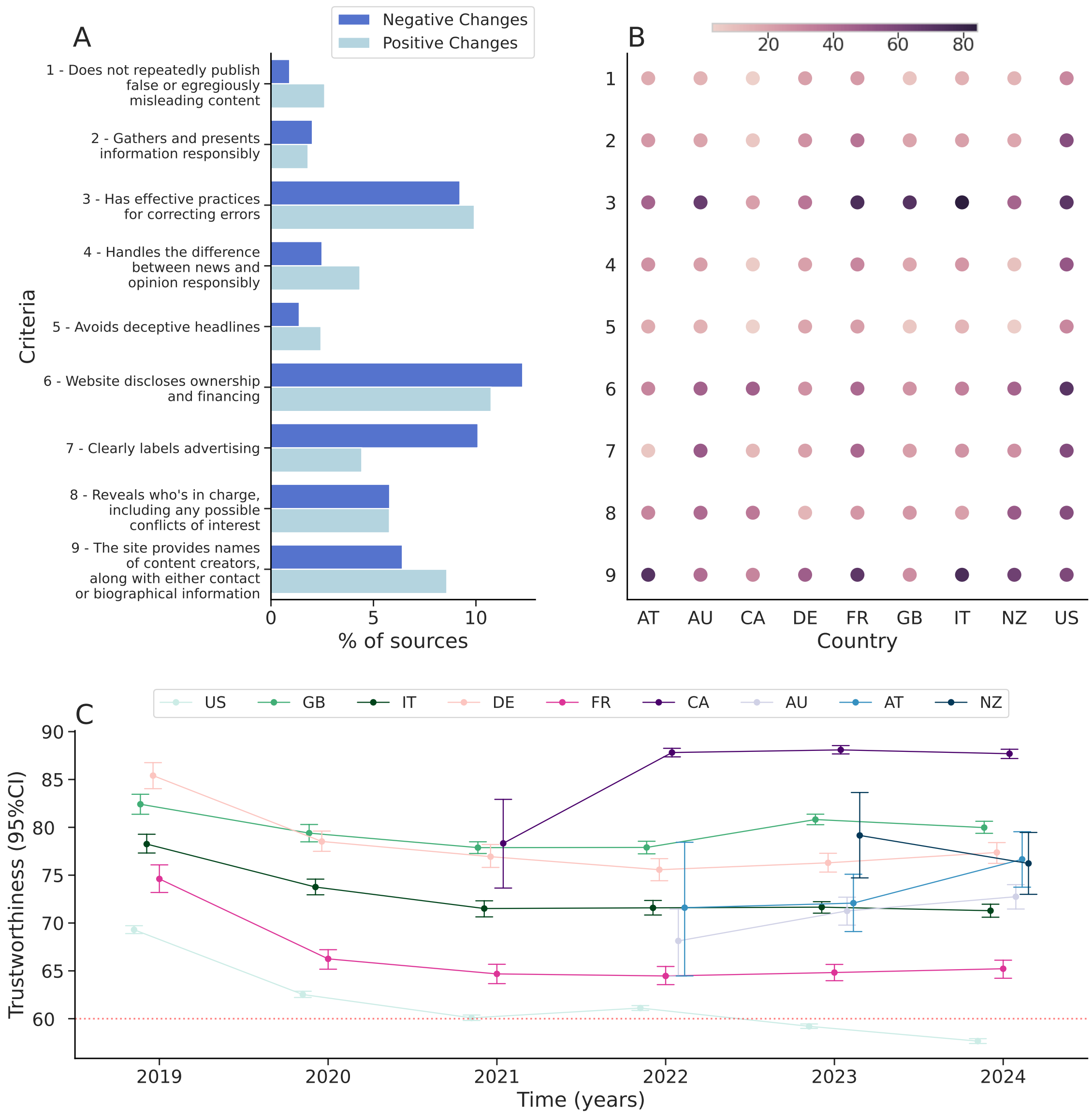

Figure 2. Panel A: percentage of sources that have stopped or started to fulfill a criterion, including multiple changes (positive vs. negative) of a single source. Panel B: percentage of sources that do not fulfill a criterion per country (July 15th, 2024). Panel C: trustworthiness per country over time (truncated to range from 55 to 90), aggregated per year with 95% confidence intervals.

Reproducibility of downstream research results using different snapshots of the database. Over the observation period, we observe both a substantial addition of sources, as well as more minor changes in individual journalistic criteria and therefore, the overall score for individual sources. This begets the question of whether such changes can potentially influence the conclusions drawn from investigations that rely on the NewsGuard database, depending on which version is used. To provide a partial answer to this question, we reproduce some of the analyses of the trustworthiness ratings of sources shared by politicians in the US, UK and Germany by Lasser et al. (2022), using different versions of the NewsGuard database. In Figure 3A, we show the temporal development of the average NewsGuard score of sources shared by Democrat and Republican members of Congress on Twitter between 2016 and 2022. We use snapshots of the NewsGuard database taken on March 1 from 2019 to 2024 (the original research used the snapshot from March 1, 2022). For Republicans, the result barely changes between different versions of the database, except in 2019, where scores are substantially higher–on average 3.2 points higher than the snapshot from 2020. The differences between snapshots from other years do not surpass 0.3 points and reveal no trend. For Democrats, the snapshot from 2019 also has substantially higher scores (2.1 points when compared to 2020). In addition, the snapshot from 2024 yields a 1.3 points lower average score than the snapshot from 2023. For other years, the differences also do not surpass 0.3 points and have no trend. However, even given the sometimes relevant differences in scores, the main finding of the research, namely the increasing difference in average scores between Democrats and Republicans, does not change. Figure 3B shows the average difference in the NewsGuard score of sources shared by the politicians over time. When using the 2019 snapshot, which deviates most strongly from the others, the difference between parties is smaller in the years 2020 to 2022 compared to other snapshots, but remains substantial.

The finding that there are only minor changes depending on which version of the database is used for the US is likely due to the fact that sources rarely change their rating, and that almost all major US news sources were already covered in the first version of the NewsGuard database: the snapshot from 2019 covers 15.0% of links posted by Democrats on Twitter and 18.5% of links posted by Republicans. For the 2024 snapshot, the coverage slightly increases to 16.3% and 19.6% of links for Democrats and Republicans, respectively.

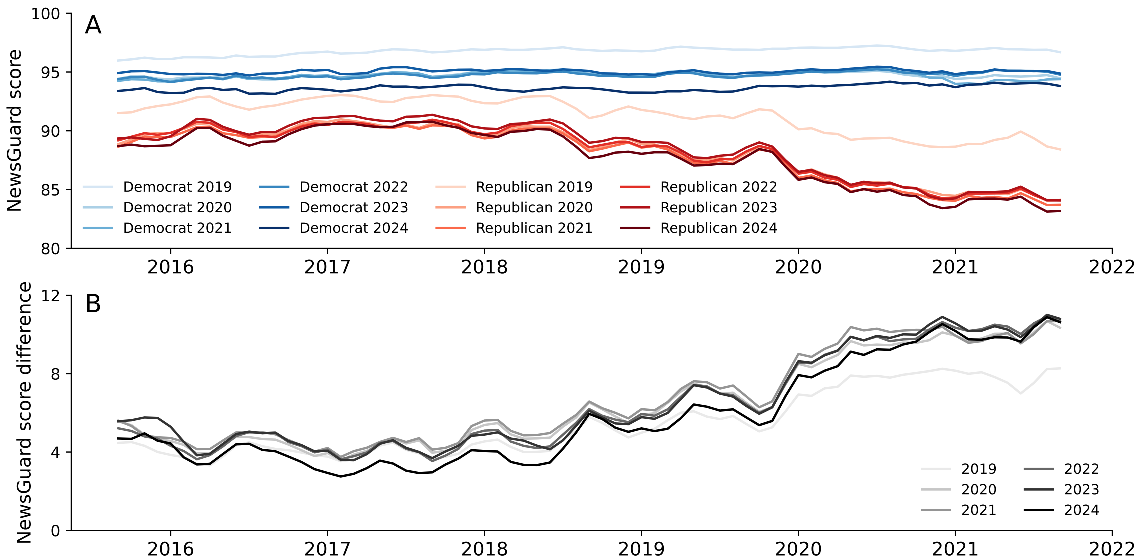

Figure 3. Reproduction of the time series of NewsGuard scores of US Congress Members from Ref. Lasser et al. (2022) using different versions of the NewsGuard database from 2019 to 2024 (always taken from the first hour of March 1st of the given year). Panel A: average scores for Democrats and Republicans indicated in different shades of blue and red for each year. Panel B: difference between the average Democrat and average Republican score, indicated in different shades of grey for each year. The time series represent a moving average over three months.

For the other two countries, Germany and the UK, the picture changes: differences in average ratings stratified by party show larger differences between years (see Figures A.3 and A.4 in the appendix). This is likely due to massive changes in the coverage of the sources shared by politicians in these countries by NewsGuard. For example, for members of the German party SPD, coverage increased from 0.5% in 2019 to 12.7% in 2024 and from 0.2% to 12.0% for members of CDU/CSU. In the UK, the coverage for members of the Labour Party increased from 7.8% to 13.5%, and for Tories, from 2.4% to 7.8%. Taken together, such coverage differences between years would lead to different conclusions in some cases, but not in others. For instance, the NewsGuard score of small parties like DIE LINKE and the Greens in Germany shows an upward trend with the 2019 version but not with newer versions. In the UK, all versions would yield the same main conclusion, namely that the average score shows no specific trend over time. It is usually the earlier versions of the database that lead to different scores, given that coverage was still low in 2019. Later versions, especially 2022-2024, usually yield the same result.

This analysis shows that for countries other than the US, the usage of different versions of the database has the potential to significantly influence research outcomes, likely due to the addition of a significant number of relevant new sources to the database over the years. Researchers investigating countries for which sources were only recently added to the database (e.g., New Zealand or Austria) should, therefore, proceed with caution and invest time to validate the coverage of the database for a given country.

How complete is the database across countries, particularly for non-US countries?

Country coverage. The dataset (2019-2024) contains sources from nine countries in total (United States, Great Britain, Italy, Canada, France, Germany, Austria, Australia, and New Zealand) plus sources considered to be global. Countries were added in the following order: Great Britain (country name in the database "GB"), United States (US), Italy (IT), France (FR), Germany (DE) – all added in 2019, Canada (CA, added in 2021), Australia (AU, added in 2021), Austria (AT, added in 2022), and New Zealand (NZ, added in 2023). In the following analysis, we will drop the category "ALL".

The majority of sources are from the United States (76.1%), followed by Great Britan (5.2%). Table 1 gives an overview of the total number of sources per country and their average trustworthiness scores (also see Fig. A.5 for the number of sources over time per country). NewsGuard also labels the language of a source, with English being the most represented (9,455 sources; 86.9%), followed by Italian (4.9%), French (4.4%), and German (3.8%).

Table 1: Country descriptives.

Country	n (%)	Trustworthiness (SD)	Updated	Changed
US	8281 (76.1)	56.8 (34.1)	224 days	559 days
GB	568 (5.2)	78.5 (23.5)	224 days	474 days
IT	537 (4.9)	70.9 (24.2)	291 days	661 days
CA	471 (4.3)	87.7 (15.3)	306 days	430 days
FR	424 (3.9)	65.4 (29.2)	326 days	646 days
DE	368 (3.4)	77.0 (31.2)	308 days	673 days
AU	166 (1.5)	72.5 (24.6)	331 days	358 days
AT	42 (0.4)	77.2 (29.6)	338 days	375 days
NZ	22 (0.2)	76.3 (24.7)	313 days	426 days

Note. As of September 15th, 2024. Sorted by size.

Across all versions of the database and in most countries, the overall trustworthiness of sources remains above 65, except for the United States. US-based outlets are of lower trustworthiness on average (see Table 1). Figure 2C shows the average trustworthiness of sources per country over time (see also Figures A.6 and A.7 for plots by country). The trustworthiness score for the US was lower from the first database version onward. It is unclear whether US sources are of lower trustworthiness overall or if larger coverage of low trustworthy outlets from the start can explain this finding. The development of trustworthiness indicates that for countries with a higher number of sources, the database seems to have settled into a stable state, whereas for newer countries (Canada, New Zealand, Australia), rating coverage still seems to vary.

When comparing the number of sources that meet the journalistic criteria per country, we observe some striking patterns. Figure 2B shows the percentage of sources that *do not fulfill* a given indicator by country, i.e., a high percentage indicates that many sources fail to meet the criterion. For most countries, more than 50% of sources meet each criterion. Interestingly, over 70% of Italian sources do not provide names of content creators, and over 80% do not effectively correct errors. A similar but less pronounced pattern is visible for French sources. Across all criteria, the United States shows the highest percentage of sources not fulfilling the criteria, with 56.4% of sources failing to gather and present information responsibly, almost twice the share of sources than in any other country. Furthermore, 30% of US-based sources fail to fulfill the heavily-weighted criterion “Does not repeatedly publish false or egregiously misleading content”.

Intervals between updates of sources vary by country (see Table 1), ranging from 224 days for the US and 291 days for Italy to 308 days for Germany. Sources get updated once a year per country, on average. Overall, the US coverage seems the most stable and complete. However, the number of media outlets in each country may depend on numerous factors (e.g., population size, fragmentation of the media system, political system, and so on). To better understand how complete the database is per country, we describe comparisons with domains shared on social media and with other lists in the following sections.

Comparison with domains shared on social media. In the supplementary material of Lasser et al. (2022), the authors describe a manual inspection of web domains in their Twitter (now X) dataset that were not covered by NewsGuard (see “Reproducibility of downstream research results using different snapshots of the database”). More precisely, the authors investigate all links posted by politicians that point to web pages other than social media platforms and search platforms. Overall, NewsGuard seems to cover a notable fraction of those links (US: 46.5%, Germany: 58.8%, UK: 39.2%). Per country, the authors selected relevant domains not covered by NewsGuard that individually accounted for at least 0.1% of all linked domains, i.e., these domains had likely been shared repeatedly. Subsequently, they inspected the missing domains and manually assigned labels to them (e.g., “government”, “news”, or “blog”).

For the US, 47 domains were analyzed, accounting for 21.2% of missing domains in the dataset. While most of these domains link to government or personal websites, none of the 47 domains lead to a news site. For the UK, out of the 78 analyzed domains (40.3% of missing domains), four domains not covered by NewsGuard are news sites, comprising a total of 1.5% of the links shared by the politicians. Of these sites, one was a clearly right-leaning site and two were mostly linked to by members of the labour party, while the last site was a news aggregator with no clear political leaning. Similarly, for Germany only a fraction of links not covered by NewsGuard lead to existing news sites (1.2%), corresponding to four out of the 62 manually analyzed missing domains (31.5% of missing domains analyzed). These sites were primarily shared by members of the left, green, or liberal parties, indicating that they are not widely shared across party affiliations.

Although the investigation examines only 20-40% of domains not evaluated by NewsGuard, by analyzing the domains that were most frequently shared, it demonstrates that NewsGuard provides a comprehensive list of domains shared by politicians in their social media communication across the US, UK, and Germany. Moreover, the small number of missing news sites and the relative absence of sources from both the left and right political spectrum in the missing domains suggests that NewsGuard's domain curation shows no systematic political bias.

Comparison with other lists. In order to assess the actual coverage of sources per country, we compare the sources covered by NewsGuard with other lists. However, other lists are often not compiled with the same objective as NewsGuard, which is to cover all sources responsible for most of the internet traffic. Therefore, their scope and rating system may differ. In addition, other lists are not available for every country.

Primarily for the US context, Lin et al. (2023) have compared the overlap and agreement between ratings across six lists of existing expert ratings, including the NewsGuard database. While trustworthiness ratings correlated significantly for the web domains included in multiple lists, the authors found a high number of non-overlapping web domains between lists.

For the German context, Puschmann et al. (2024) collected a list of 1,147 unique German online news domains (GOND)[6], largely relying on web tracking data of a representative sample of 1,500 German citizens. Overall, GOND has 552 sources in common with NewsGuard (see Table A.2, among others labeled as "legacy press" (52.9%), "digital-born news outlet" (12.0%), or "hyperpartisan news" (10.1%). NewsGuard covers 244 out of 573 German-speaking sources covered by GOND, including some other languages.

[6]https://osf.io/s5uhb/

Overall, comparisons of NewsGuard with the two lists for the US and Germany reveal many non-overlapping domains. However, NewsGuard's coverage of the United States and Germany seems extensive in that NewsGuard generally tends to cover more prominent outlets and excludes niche and regional outlets, probably because it focuses on websites with the most engagement.

How valuable are contextual source labels for misinformation research?

Political orientation. In addition to trustworthiness ratings, the NewsGuard database also classifies sources according to their political orientation, relying on a left-to-right conception of the political spectrum. In December 2022, the categories were condensed from four (far left/ slightly left/slightly right/far right) to two (left/right; see Fig. 4A). Overall, the political orientation label is only available for a minority of the sources, specifically 33.4% (3,789 sources) in the current database version.

In the US, 34.8% of news outlets have a political orientation label, in contrast to only a small minority of sources in other countries: 10.4% of sources for Great Britain, 13.2% for Italy, 16.9% for Germany, and 20.3% for France. Figure 4C and Table A.3 show the total number of political orientation labels per country as well as the number and percentage of sources classified as left or right-leaning in the most recent database. In most countries, the majority of sources with political orientation labels belong to the right-wing spectrum. Figure A.8 shows the distribution of political orientation per country over time.

Trustworthiness ratings are generally lower for right-leaning sources (see Fig. 4B). In the database snapshot from September 2024, the average trustworthiness score for right-leaning sources is 26.4 (SD=20.3), in contrast to 60.6 (SD=26.1) for left-leaning sources (M_{diff}=34.2, t(3771)=35.4, p<.001). When political orientation was still divided into four categories, slightly left- and right-leaning sources had moderate trustworthiness ratings (for example, in June 2021, 81.3 and 59.4, respectively), and far-left and far-right sources had comparatively lower ratings (54.0 and 15.9, respectively). In August 2021, however, when a number of untrustworthy sources (M=40.5, SD=37.2) were added to the database, the average trustworthiness of sources categorized as slightly right-leaning dropped by almost 20 points. Figure 4A shows that the "Slightly Right" category dropped below the "Far

Left" category by almost 15 points.

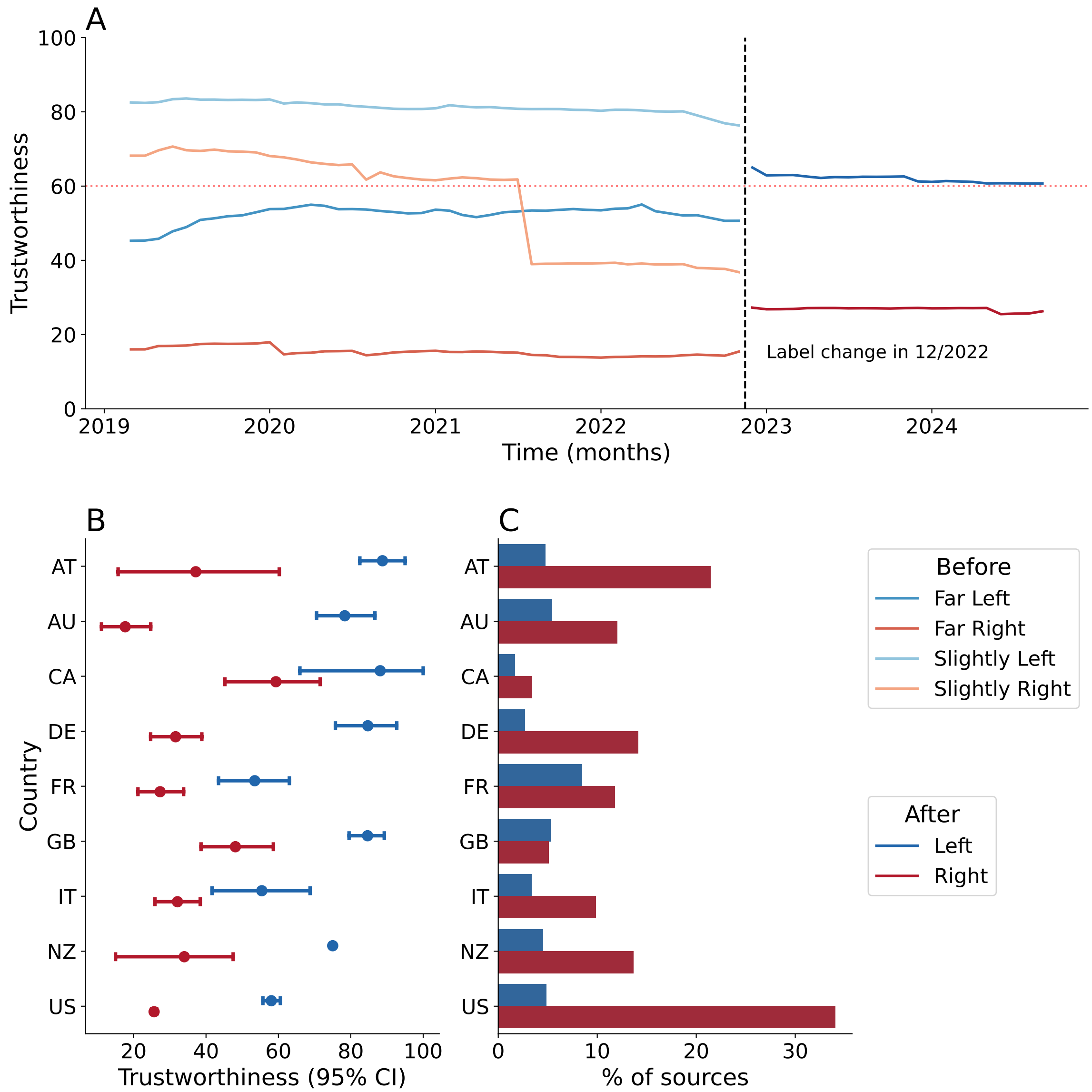

Figure 4. Trustworthiness averages by political orientation. Panel A: over time. Panel B: by country. Panel C: percentage of sources per country with political orientation, explaining differences in trustworthiness by country.

In addition to the change in average trustworthiness scores within political orientation categories caused by the changed taxonomy of political orientation used by NewsGuard, the trustworthiness averages per political orientation also differ by country, as shown in Fig. 4B. While for some countries like France and Canada, the average trustworthiness of left and right sources only differs by about 20 points and 95% confidence intervals overlap, other countries like Australia, Germany, and Austria show extreme differences of over 40 points. Nevertheless, across all countries, sources classified as right, on average, always have a lower trustworthiness score than sources on the left. However, given the small percentage of sources classified as belonging to either side of the spectrum, we cannot infer that sources on the right, in general, have lower trustworthiness. A bias in NewsGuard's procedure of classifying sources as either left or right or not at all could also produce these findings.

To get an idea of whether sources with labels are correctly classified and whether sources without labels are neutral or not classified by NewsGuard for other reasons, we compared NewsGuard ratings with manual and other expert source labels. First, we manually annotated the political orientation of all German-speaking sources ever covered by NewsGuard (as of September 15th, 2024) based on the website's front page, article headlines, ads, and article content. We followed the labels by NewsGuard (right/left) and added two labels: neutral and unidentifiable (e.g., when the website is unreachable). For sources with available ratings, the agreement between NewsGuard and the manual annotations was 83% (for 64 domains out of 406, excluding 20 dead links). Second, we compared the NewsGuard labels to labels by MediaBiasFactCheck[7] (MBFC, 41 labels in total, six overlapping with NewsGuard), resulting in an agreement of 83%. Both comparisons showed that disagreements mainly occurred in cases where the manual annotations were labeled as neutral, while the expert ratings assigned a specific label. In only one case, NewsGuard differed from both MBFC and the manual annotation.

Consequently, in the German context, the existing ratings largely matched manual political orientation labels. Interestingly, the rated sources tended to be niche outlets with more extreme (and thus, clear-cut) political leanings. For the neutral sources, we could not identify a clear political leaning. Such sources were typically local newspapers or magazines

[7]See here for rating system as of August 20th, 2024: cd https://mediabiasfactcheck.com/methodology/

focusing on special interest topics (e.g., technology or health). However, roughly half were legacy media or major online sources that slightly leaned towards a particular political orientation. Our findings suggest two (not necessarily mutually exclusive) implications for a missing political orientation: Either the sources have no overt political bias in their reporting or are too niche to be labeled by NewsGuard.

Before the rating became binary, a few domains frequently oscillated between more extreme and less extreme labels, but domains rarely switched their political orientation. This might have been a reason for simplifying the label. Only five domains have changed their political orientation rating from left to right (none have changed from right to left). These domains are US-based, and two of them also experienced a significant drop in their trustworthiness score, reducing it by 42 points in January 2020. We therefore conclude that political orientation labels generally remain stable for the relatively small number of sources that have such a label.

In summary, although the political orientation labels of sources provided by NewsGuard are sparse, particularly outside the US, the ratings of covered sources appear generally stable over time. An in-depth investigation of labels for German sources showed that NewsGuard's political orientation assessment is reasonable and the low number of labels can be explained by sources not showing a political leaning or being too niche. Therefore, the political orientation rating by NewsGuard may be a valuable addition to analyses related to source trustworthiness. However, we recommend that researchers (at least) spot-check the ratings for the specific country in which they intend to use the database, especially for major outlets and mainstream media sources. If the research focuses on comparing political bias rather than controlling for it, we advise validating and extending the existing ratings provided by NewsGuard.

Topics. NewsGuard experts assign labels for the topics covered by the sources. Usually, a single source has multiple assigned topics. The number of sources with topic labels has steadily increased since their introduction in October 2019. As of May 2021, roughly 50% of the sources had topic labels (see Fig. A.9) for the total number and proportion of sources with topic labels). In September 2024, only 352 sources are without a topic label (2.9%). In Figure 5B, we show the topic counts. The most popular topics are

"Local news" (40.3% of sources), "Political news or commentary" (39.5%), "General news" (22.7%), "Health or medical information" (13.6%), and "Conspiracy theories or hoaxes" (12.8%). Over time, topic labels related to misinformation have been added, most likely coinciding with some of the major database updates discussed above. Figure 5C shows the top five topics discussed by sources classified as untrustworthy (measured by the percentage of untrustworthy sources publishing on those topics). In April 2020, shortly after the COVID-19 pandemic started, NewsGuard increasingly added a label called "COVID-19 misinformation", with major additions in August 2021. Once a source receives a topic label, it rarely changes (except for foxnews.com, which has changed seven times).

As shown in Figure 5A, average trustworthiness ratings greatly differ across topics (white dots show the average trustworthiness of sources covering that topic). The lowest average rating is given to sources covering issues related to conspiracy theories (M=12.6, SD=12.6), military (M=31.8, SD=34.9), and health or medical information (M=39.3, SD=35.5). In contrast, those covering education have the highest average trustworthiness scores (M=85.8, SD=16.9, but this is only 0.5% of the sources). Generally, topics with low coverage tend to have high trustworthiness scores with low variance (e.g., fashion with 76.5 on average, SD=10.2), see Figure 5B. Considering only untrustworthy sources, however, only 32.5% of them are labeled as covering conspiracy theories. Instead, untrustworthy sources seem to commonly publish "Political news or commentary", with 79% of all untrustworthy sources (e.g., with a trustworthiness score $<$ 60) covering this topic, followed by local (30.4%) and health news (22.3%). These topics generally receive a significantly higher number of labels from NewsGuard and show a high variance in trustworthiness, for instance, "General news" or "Local news" (see Fig. 5A and B). This highlights that both untrustworthy and trustworthy sources cover a wide range of topics, including mainstream ones, and that fringe topics only make up a small proportion of the news.

Across countries, sources more or less publish on the same topics (see Fig. A.10 in the appendix). However, sources with low trustworthiness seem to publish on different topics, i.e., the trustworthiness of sources covering a given topic substantially depends on the country (see Fig. A.11). For instance, for medical information, sources in Canada are rated as trustworthy (M=65.4, SD=34, 27 sources), while trustworthiness is well below 60 in the United States (M=30.2, SD=33.3, 1,130 sources). The difference between countries

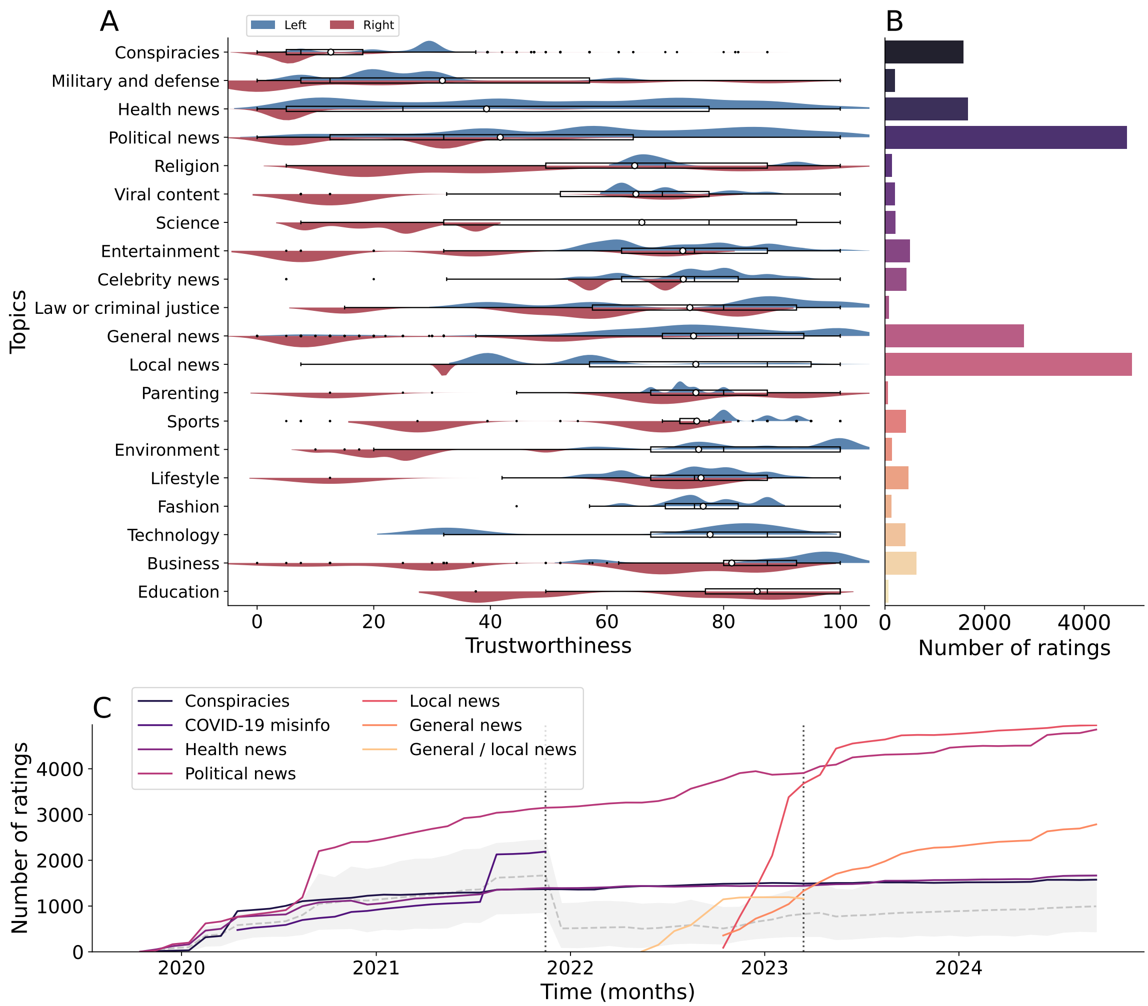

Figure 5. A) shows the distributions of trustworthiness per topic and political orientation (left/right in blue/red, respectively), sorted by their average trustworthiness (white dot within the boxplot that represents the quartiles and whiskers show the whole range). Note: We excluded topics with a count below 50. Some topics are abbreviated. B) gives the count of topics as of September 15th, 2024. C) shows the frequency of misinformation-related labels over time. Colored lines show the top five topics covered by untrustworthy sources. The dotted, vertical lines show the removal of two of those labels. The grey and dashed line shows the average topic label count of those topics across all sources with the 95%CI.

is also substantial for political news, with an average trustworthiness of 33.1 (SD=24.8, 3,498 sources) in the US, 39.2 (SD=29.3, 111 sources) in France, and 81.6 (SD=18.6, 268 sources) in Great Britain.

We also analyzed the interaction of average trustworthiness by topic and political orientation (see Figure 5A). Political and local news are among the most popular topics assigned to both left- and right-wing sources, with 386 and 113 as well as 2,946 and 1,140 sources covering those topics, respectively. These numbers show the differences in how many political orientation labels are assigned to each side. For sources classified as right-wing, trustworthiness scores for all topics are consistently lower on average than for sources on the left side, e.g., political news: M=26.6 (SD=20.0) on the right and M=60.2 (SD=26.0) on the left. 1,162 right-wing sources cover conspiracy theories (M=10.0, SD=11.4) and 693 sources cover health-related news (M=11.3, SD=15.0). Meanwhile, for sources labeled as left-leaning, trustworthiness is particularly low (below 50) when they cover the topics "Conspiracies", "Military and defense" or "Health news". However, due to the skewed distribution of political orientation in the database, it is impossible to say whether the above-mentioned differences in countries and political orientation are due to actual differences in the media landscape or a sampling bias.

Overall, the coverage of topic labels appears comprehensive, suggesting they could be a valuable asset for misinformation research. Especially a combination of trustworthiness and topic labels could complement source characterization, for instance, by distinguishing between fringe sources and those addressing mainstream issues. For sources that lack a label for political orientation, topics may help clarify if a source is genuinely neutral or tends to cover extreme topics low in trustworthiness.

Conclusion

An inherent challenge in misinformation research lies in how false information is defined and measured. A prominent approach for tracking online misinformation is to use a list of sources and their web domains. Due to its popularity and limited access, we examined the comprehensive NewsGuard database and analyzed the temporal stability and cross-country completeness of their trustworthiness ratings and other source labels relevant to misinformation research. Here, we summarize practical recommendations for using NewsGuard in research and discuss conclusions for source-based approaches in general.

Over 50% of the sources included in the database have a score of 60 points or higher, deemed at least "credible with exceptions" according to NewsGuard's nomenclature. These sources typically meet two high-weight criteria: "Avoids deceptive headlines" and "Does not repeatedly publish false or egregiously misleading content". The average trustworthiness score of sources included in the database has decreased over time due to the addition of untrustworthy sources rather than the degradation of trustworthiness of existing ones. NewsGuard provides a new snapshot of its database hourly, but an analysis of update time stamps of sources shows that source information is likely checked annually, with infrequent changes to trustworthiness ratings usually triggered by changes in transparency about website ownership and financing or editorial practices. Such changes are generally minor, involving updates of only one of the nine journalistic quality criteria. However, changes in source coverage can be abrupt, with instances of over 1,000 sources being added or removed in a single update. To shed some light on the impact of changes in the NewsGuard database on downstream research outcomes, we reproduced research investigating news-sharing practices of political elites in the US, Germany, and the UK (Lasser et al., 2022). We find that after 2019, the content of the database largely reached a stable state in the US, and using different versions of the database between 2020 and 2024 does not change research outcomes. This process took longer for Germany and the UK, but after 2022, we also only observed minor database changes for these countries. Despite these nuances, the trustworthiness ratings by NewsGuard appear relatively insensitive to time, speaking for the reliability of source-based approaches.

US sources generally score lower on trustworthiness compared to other countries. Importantly, this does not say anything about the trustworthiness of news actually consumed by the US population, which could be higher, depending on the contribution of each source to the news diets of US Americans. Country-specific factors seem to significantly influence journalistic traditions across countries with similar media systems. Italy and France, part of the Southern cluster, show distinct patterns compared to Central (e.g., Germany) and Western clusters (e.g., US). According to Brüggemann et al. (2014), countries belonging to the Southern cluster have a less professionalized journalism sector than the Central or Western clusters. In their seminal definition of media systems, Hallin & Mancini (2004) state that in countries that fail to meet the professionalism indicators, the understanding of journalists serving the public interest is less pronounced. In other words, when journalistic professionalism is low, journalists lack professional autonomy, potentially explaining less institutionalized transparency practices in these countries, as indicated by NewsGuards' journalistic criteria. NewsGuard's criteria broadly reflect differences in media systems—at least for the Western, educated, industrialized, rich, and democratic countries covered. Nonetheless, the differences in journalistic traditions underscore that applications of the database need to be tailored to individual contexts and that researchers should double-check the relevance of the criteria for the country studied[8].

The NewsGuard database provides more than just trustworthiness ratings; it also includes labels for the political orientation of sources and covered topics, which could be valuable contextual information for misinformation research. While most sources have multiple topic labels, only a minority of sources have one for political orientation. While about 41% of sources have a political orientation label in the US, only around 10% do in other countries. In our manual analysis of German sources, we observed that NewsGuard tends to rate only sources with a clear political leaning. Most rated sources are right-leaning, and these are often deemed less trustworthy compared to left-leaning sources. This disparity raises the question of whether it stems from a sampling bias by NewsGuard editors rather than an actual difference in trustworthiness. We argue that NewsGuard seems unbiased for the countries where coverage and ratings have stabilized for the following reasons: First, NewsGuard selects sources based on web tracking data, suggesting that it should reflect the online media landscape for the respective country. Research on alternative media suggests the emergence of right-wing counter-publics as a response to the political and legacy media context (Heft et al., 2020). More precisely, the theory posits that a prevalence of left-leaning opinions in political and mainstream media spheres (Osmundsen et al., 2021) may lead to an under-representation of right-wing views and the fragmentation of a counter-public along the right-wing spectrum that is highly connected (Heft et al., 2021). As NewsGuard only labels the political orientation of explicitly partisan, sometimes hyper-partisan sources, this explanation would result in a higher number of smaller, right-wing online sources. A demand for right-wing fringe publications would also explain lower trustworthiness on the right-wing spectrum, which aligns with the political asymmetry repeatedly observed in susceptibility to misinformation, e.g., recently by Robertson et al. (2023). Second, the source

[8]https://www.newsguardtech.com/ratings/rating-process-criteria/

selection and trustworthiness ratings provided by NewsGuard are largely consistent with other independent fact-checking organizations (Lin et al., 2023). Therefore, it is unlikely that NewsGuard has an inherent bias against right-leaning sources, both in selecting more right-wing sources and in giving them lower trustworthiness ratings. NewsGuard also discusses this topic in their 2023 Social Impact Report[9], emphasizing that their trustworthiness ratings are intended to be apolitical. Despite being incomplete, the political orientation labels may be a useful addition to the trustworthiness ratings, especially in combination with topic labels. However, they should be interpreted with caution and potentially validated and extended for countries other than the US and Germany.

Our analysis faces several limitations due to a scarcity of comparable datasets, particularly for countries other than the United States, the United Kingdom, and Germany. As a result, we cannot assess the reliability and validity of the NewsGuard database; we can only evaluate the internal and temporal stability of ratings. To address this, we reproduced previous research findings using different database versions, compared the database with the few existing lists, and manually validated ratings. Based on these analyses, the NewsGuard database appears stable and complete for Great Britain, the US, Germany, and possibly France and Italy. Additionally, our results are reproducible across different database versions, even with slight changes in the instrument, especially for its early versions. Given our observations, we recommend that researchers using NewsGuard's trustworthiness ratings examine several snapshots of the database to ensure no major source additions or deletions occurred recently. For countries included in NewsGuard for longer, e.g., the US, Canada, France, Italy, Germany and the UK, we suggest that researchers use whatever version of the database after 2022 they have access to. Using different versions of the database for different portions of any data being analyzed from different time periods does not seem necessary, given the general stability of trustworthiness ratings once a source has been included in the database. For more recent additions, e.g., Australia, Austria and New Zealand, we suggest that researchers always use the most recent version of the database and carefully examine it for differences in coverage for different units of analysis, such as political leaning. Lastly, the topic and journalistic criteria ratings appear stable and may offer valuable context for the analysis of misinformation dynamics. A combination of political orientation and topic labels

[9]https://web.archive.org/web/20240228003711/https://www.newsguardtech.com/special-reports/social-impact-report-2023/

could effectively characterize sources and identify untrustworthy sources that address and decontextualize mainstream issues beyond hyper-partisan contexts. It is important to note that this analysis is specifically relevant to online misinformation research, as NewsGuard uses web traffic data to select sources for assessment. Future research should compare lists of online news domains to cross-platform and offline news consumption and examine how sampling based on sources versus stories influences downstream results.

In conclusion, using a list of rated online news domains appears to be a stable method for identifying misinformation online at scale. A source-based approach with a fine-grained rating scheme based on journalistic quality criteria is particularly valuable and theoretically sound because it can encompass a wide range of sources, including less extreme forms of misinformation, thereby better-reflecting people's information diets. While the NewsGuard database serves as a useful tool for tracking online misinformation at the level of the source, researchers should weigh its utility for the studied context against associated costs and consider alternative sources (e.g., Lin et al., 2023) for comprehensive analysis. Our findings underscore the critical need for dynamic, multifaceted, and openly accessible methods to get a clear and robust answer on the impact of misinformation on entire populations.

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Data and Code Availability

Reproduction materials, including code and extended data, excluding the NewsGuard database, which is proprietary, are accessible on Github. To license the NewsGuard database, contact support@newsguardtech.com.

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Online Appendix

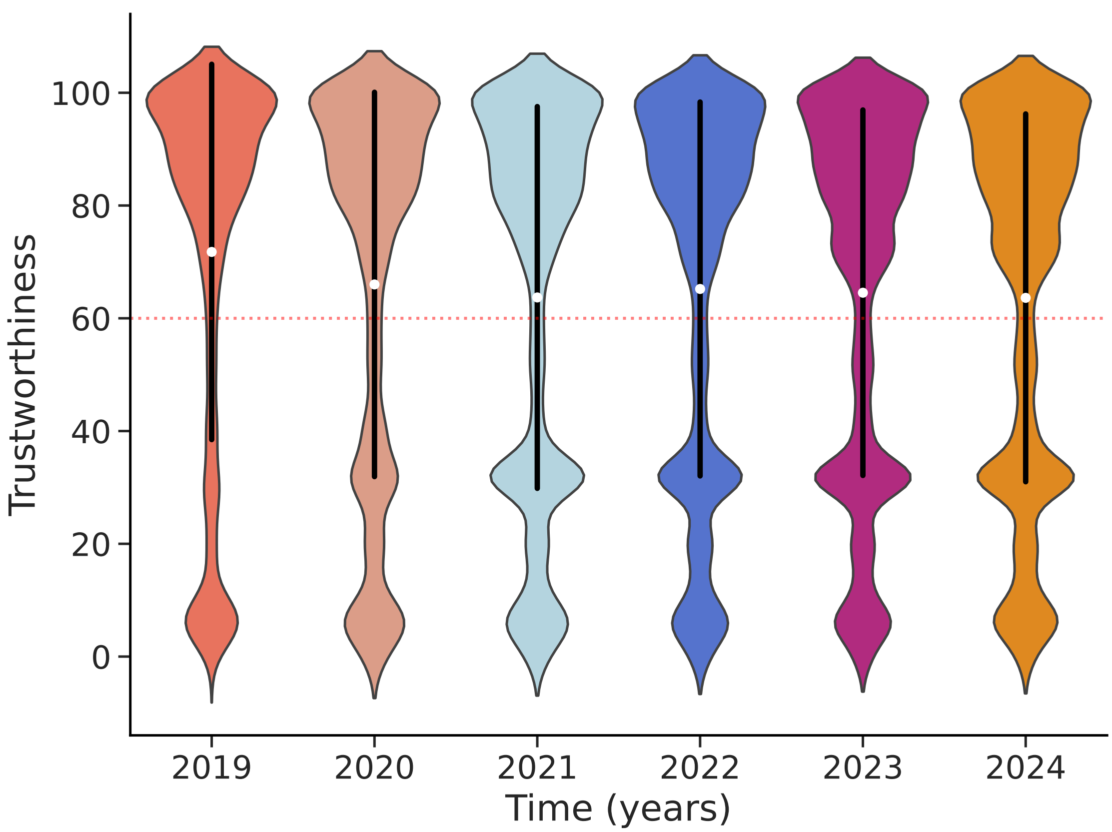

Figure A.1. Distribution of trustworthiness per year as a violin plot (with M and SD), standardized by count.

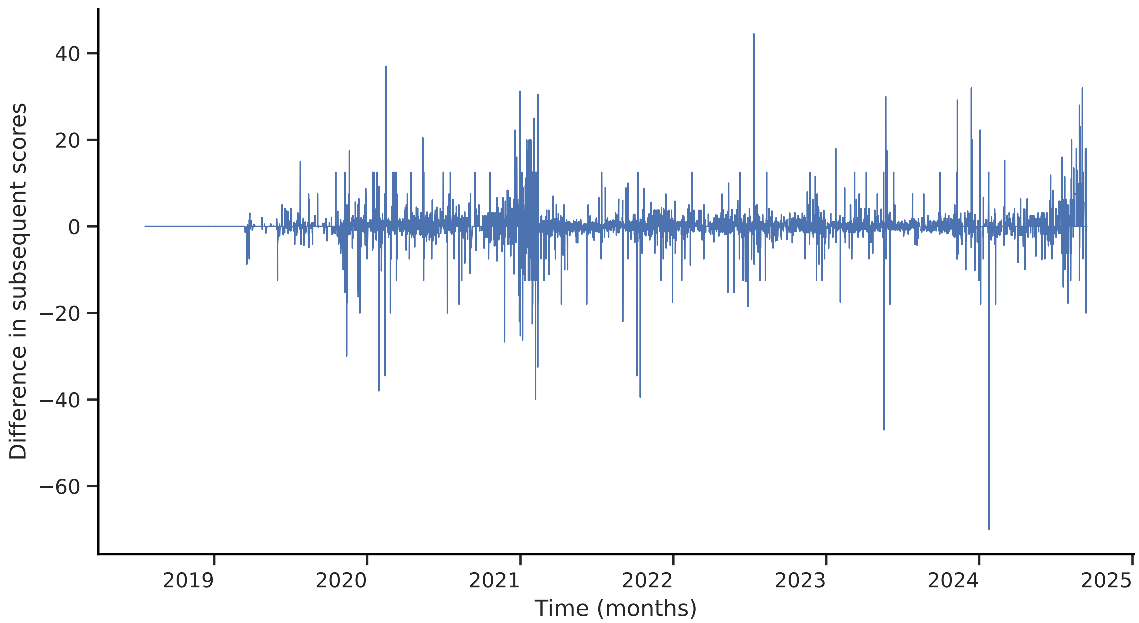

Figure A.2. Changes in trustworthiness scores over time, calculated as the difference in scores per news domain between consecutive updates.

Table A.1: Correlation matrix of journalistic quality criteria.

	1	2	3	4	5	6	7	8	9
1	1	0.63	0.4	0.56	0.93	0.38	0.39	0.26	0.36
2	0.63	1	0.57	0.75	0.63	0.58	0.49	0.55	0.56
3	0.4	0.57	1	0.48	0.4	0.45	0.44	0.34	0.53
4	0.56	0.75	0.48	1	0.57	0.52	0.51	0.47	0.5
5	0.93	0.63	0.4	0.57	1	0.39	0.37	0.28	0.39
6	0.38	0.58	0.45	0.52	0.39	1	0.42	0.48	0.4
7	0.39	0.49	0.44	0.51	0.37	0.42	1	0.29	0.36
8	0.26	0.55	0.34	0.47	0.28	0.48	0.29	1	0.47
9	0.36	0.56	0.53	0.5	0.39	0.4	0.36	0.47	1

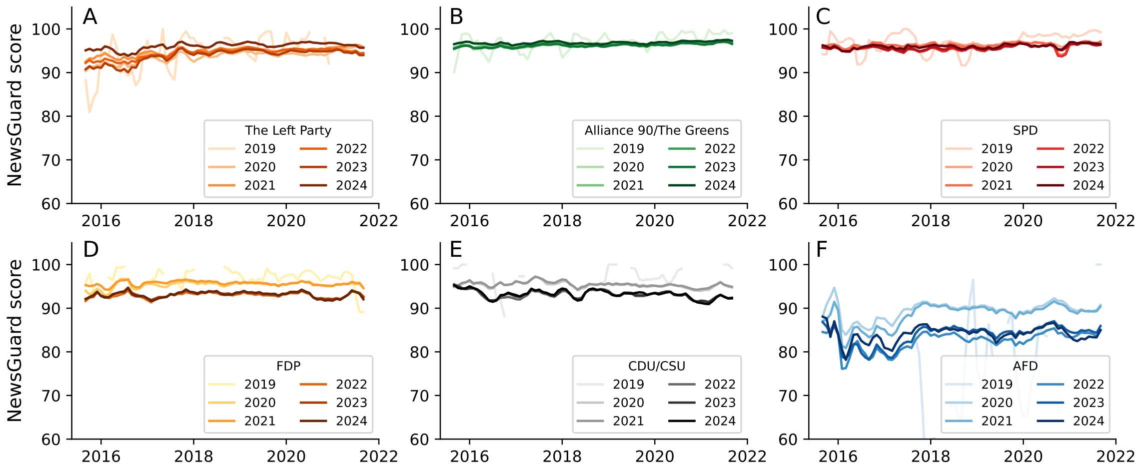

Figure A.3. Reproduction of trustworthiness ratings of sources shared by German politicians on Twitter over time from Lasser et al. (2022). Panel A shows average NewsGuard scores in Twitter posts by members of the party DIE LINKE, panel B for members of the Greens, panel C for members of the SPD, panel D for members of the FDP, panel E for members of the CDU and CSU, and panel F for members of the AFD.

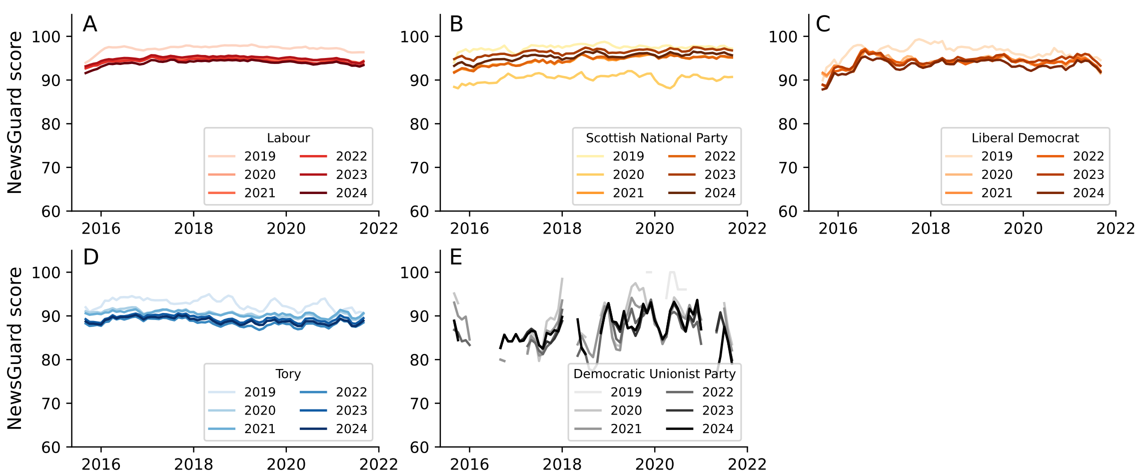

Figure A.4. Reproduction of trustworthiness ratings of sources shared by UK politicians on Twitter over time from Lasser et al. (2022). Panel A shows average NewsGuard scores in Twitter posts by members of the Labour party, panel B for members of the Scottish National Party, panel C for Liberal Democrats, panel D for Tories, and panel E for members of the Democratic Unionist Party.

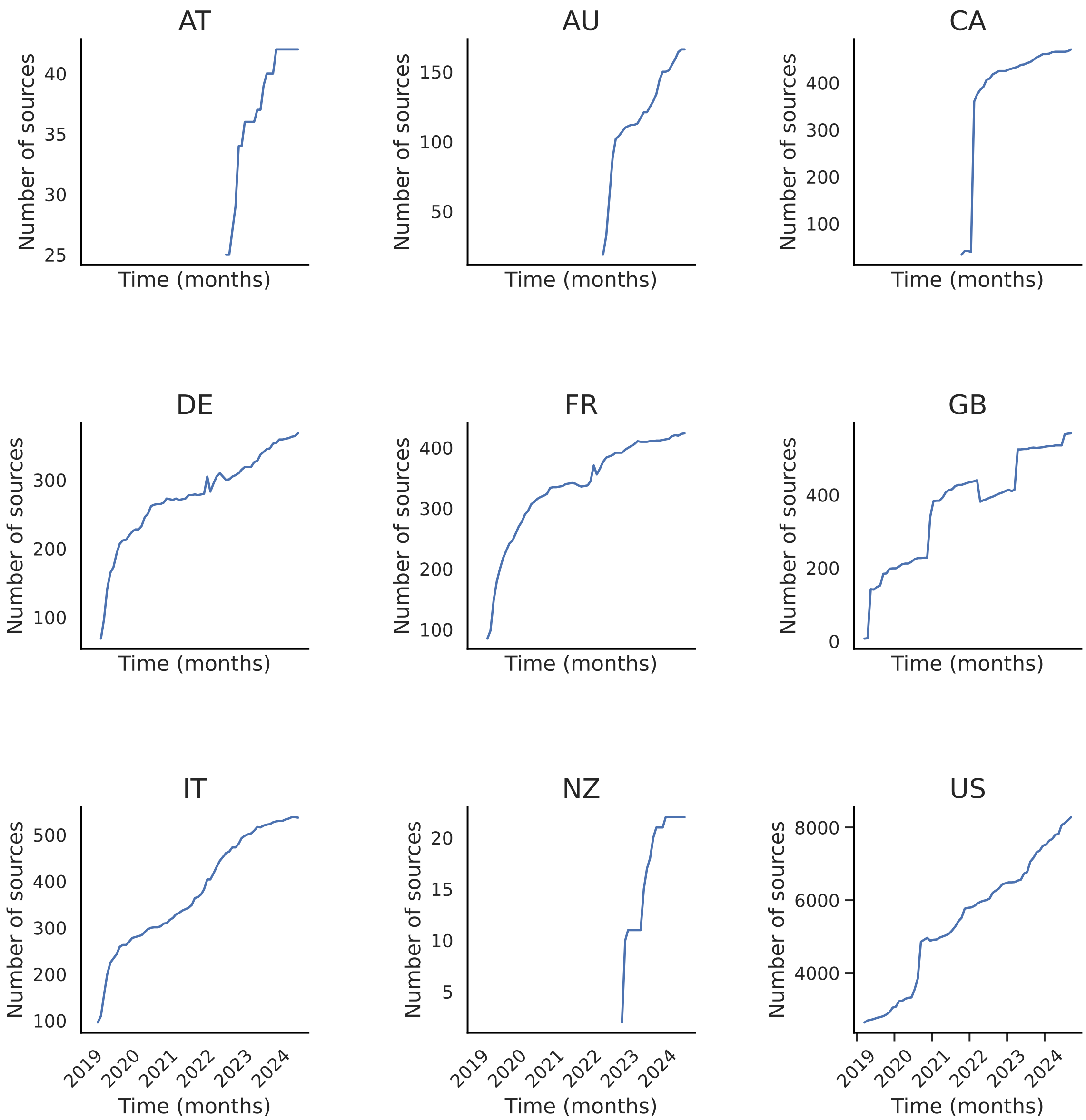

Figure A.5. Number of sources over time, per country.

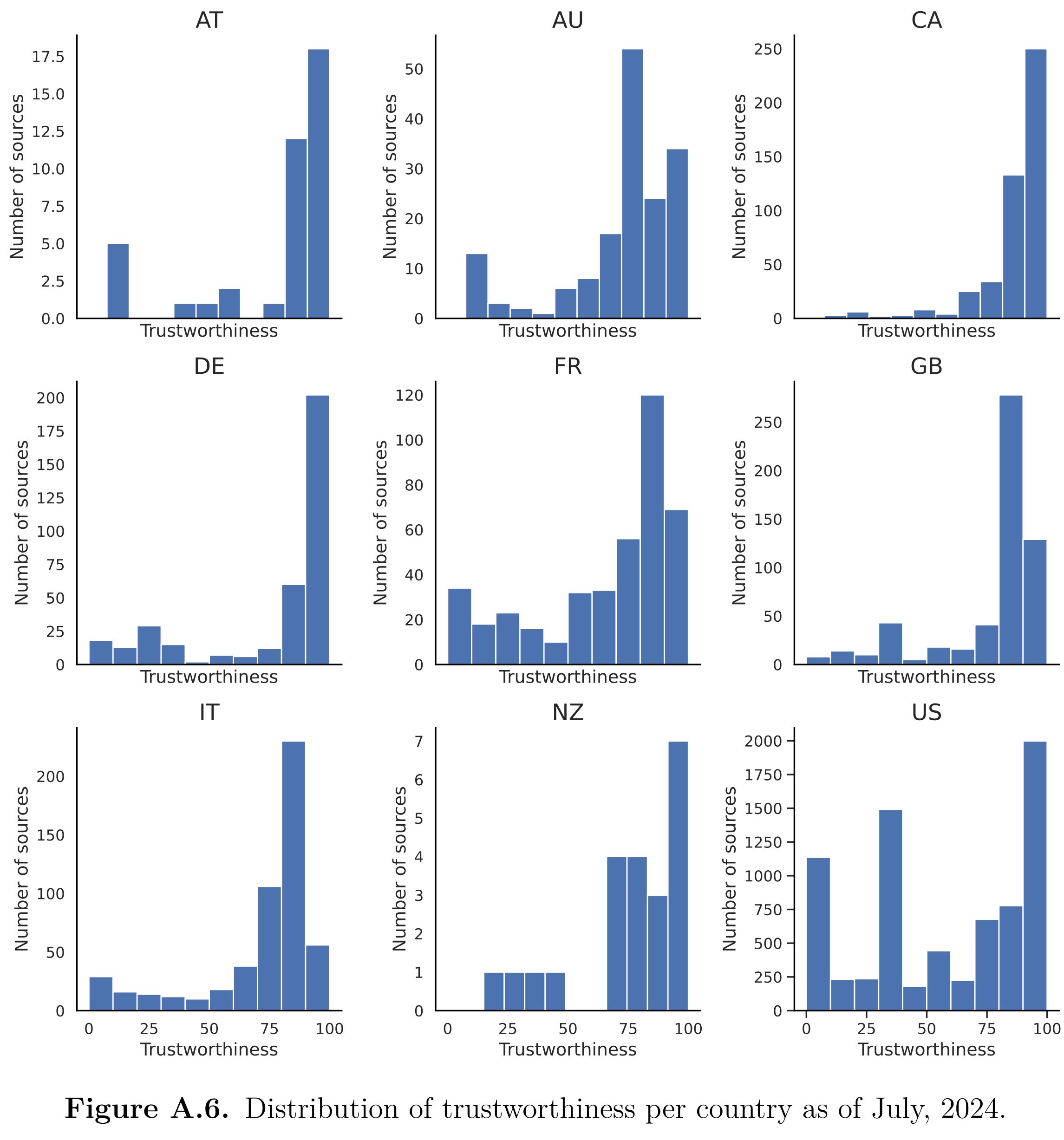

Figure A.6. Distribution of trustworthiness per country as of July, 2024.

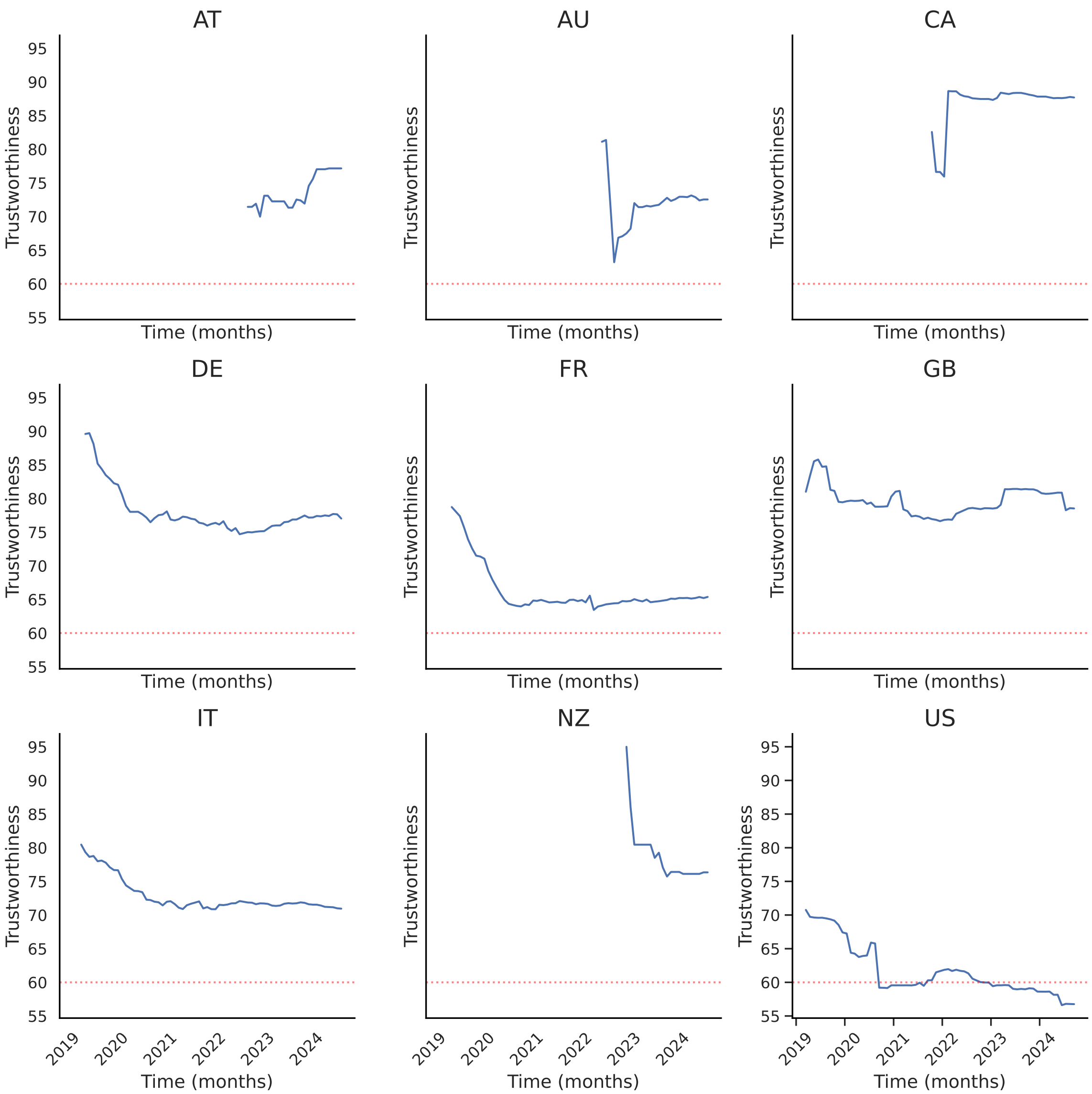

Figure A.7. Trustworthiness per country over time, aggregated by month.

Table A.2: Overlap between GOND and NewsGuard.

GOND Type	NewsGuard Score	Overlap	%
commercial broadcaster	89.4	58	10.5
digital-born news outlet	84.5	66	12.0
hyperpartisan news	43.0	56	10.1
legacy press	91.0	292	52.9
public broadcaster	89.0	40	7.2
tabloid newspaper	84.6	40	7.2

Note. As of July 15th, 2024.

Table A.3: Orientation per country.

Orientation Country	Left (%)	Right (%)	Total
US	401 (12.45)	2819 (87.55)	3220
FR	36 (41.86)	50 (58.14)	86
IT	18 (25.35)	53 (74.65)	71
DE	10 (16.13)	52 (83.87)	62
GB	30 (50.85)	29 (49.15)	59
AU	9 (31.03)	20 (68.97)	29
CA	8 (33.33)	16 (66.67)	24
AT	2 (18.18)	9 (81.82)	11
NZ	1 (25.0)	3 (75.0)	4

Note. As of September 15h, 2024.

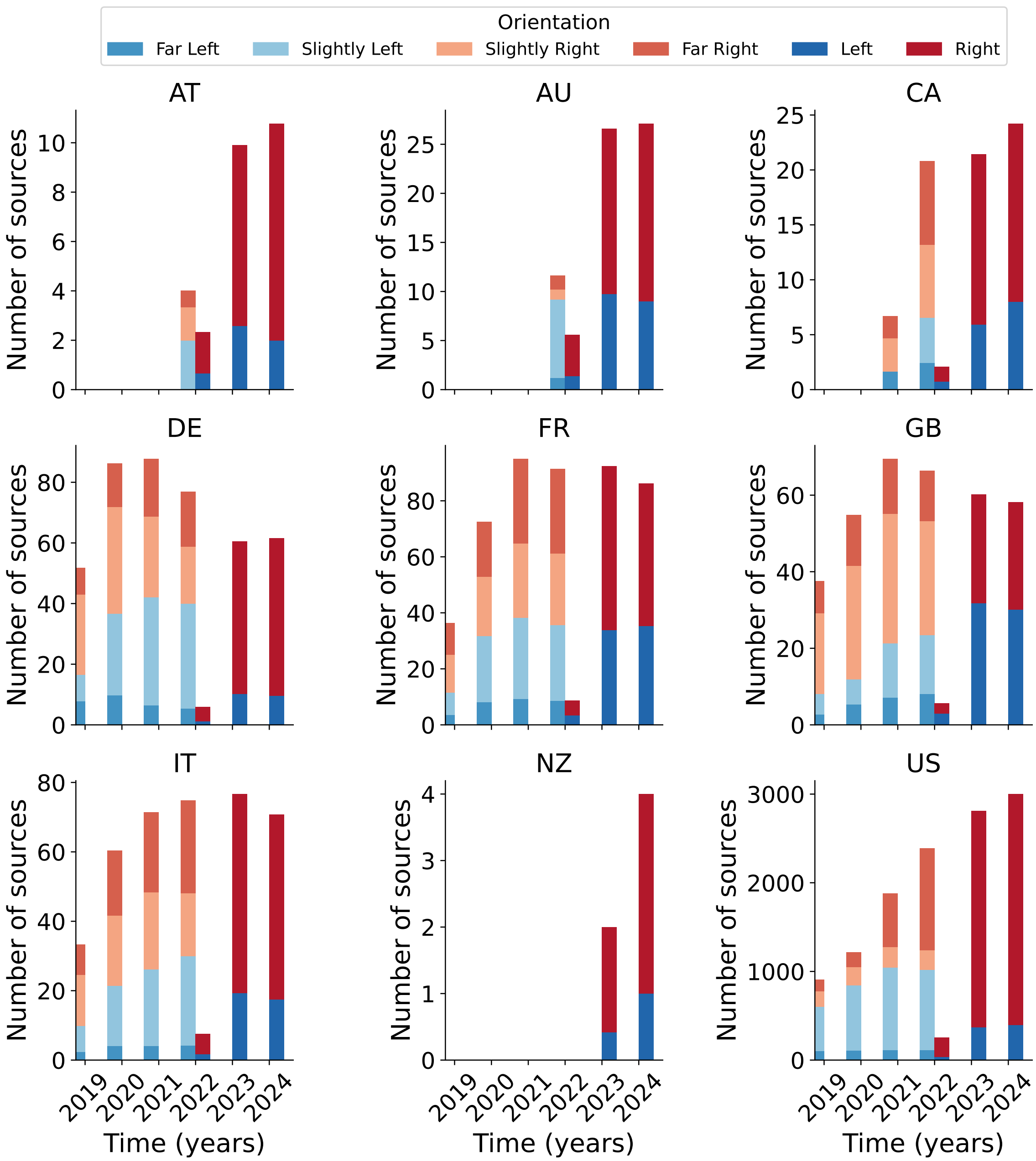

Figure A.8. Distribution of political orientation over time, by country

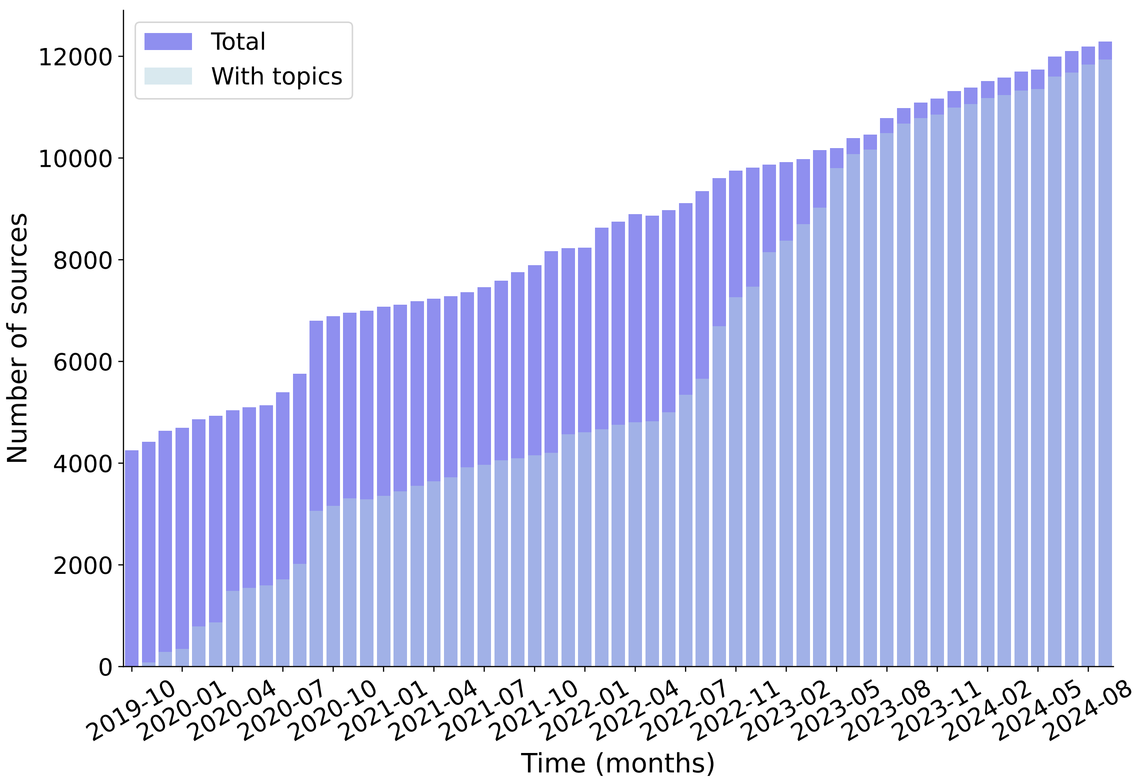

Figure A.9. Number of sources over time with and without topic labels.

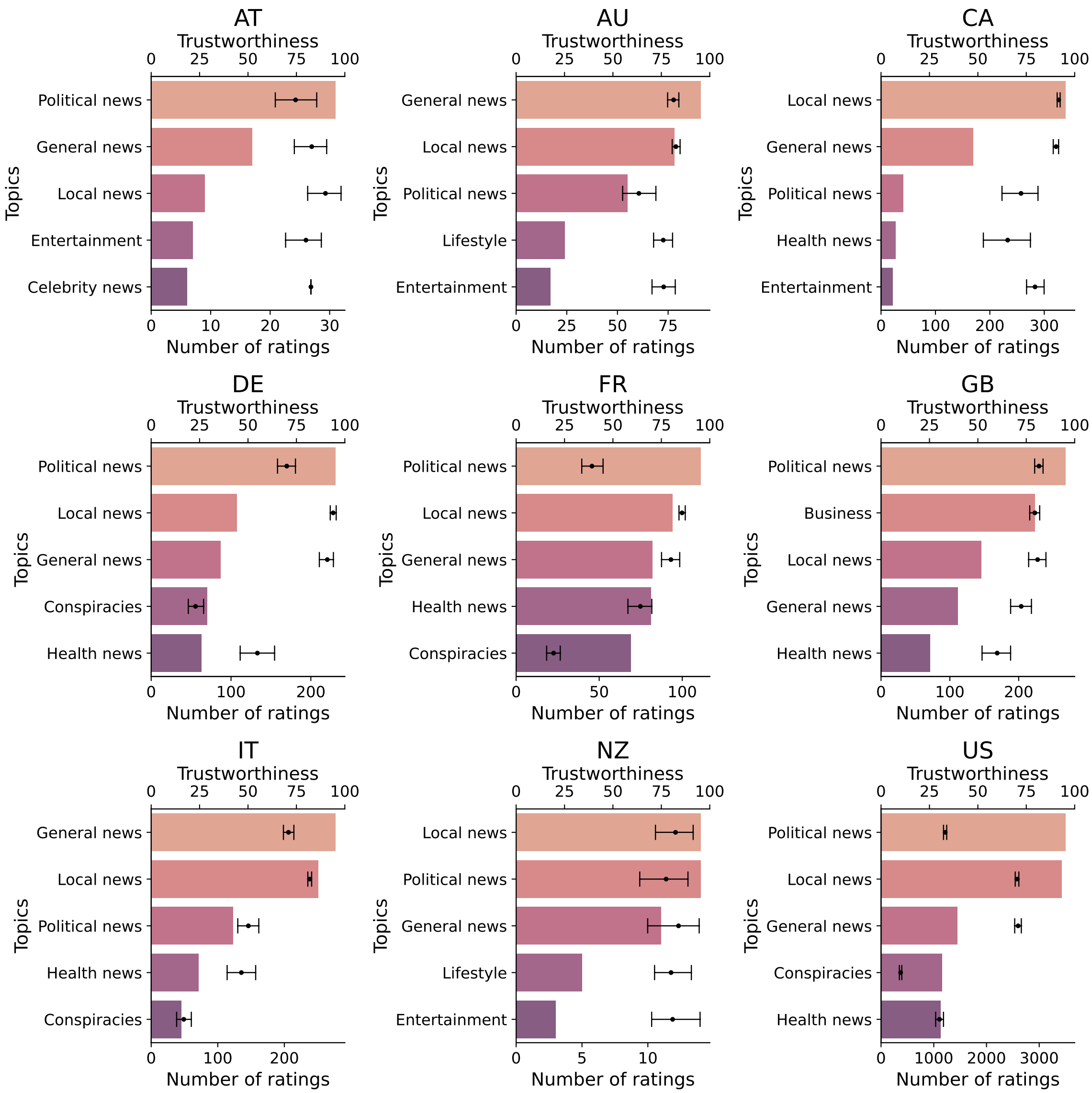

Figure A.10. Top five most popular topics per country and their total count, with the second x-axis at the top showing the average trustworthiness and standard deviations of those topics..

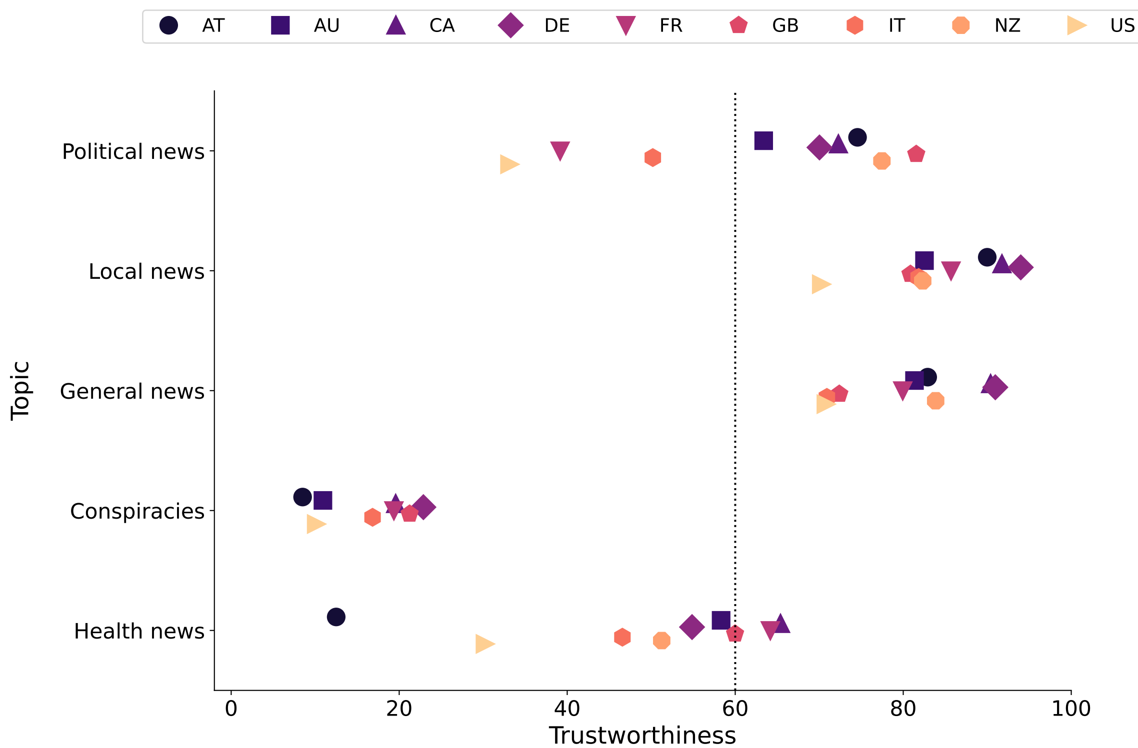

Figure A.11. Trustworthiness score for the top five most popular topics across countries in the most recent database (September 15, 2024).